

Wenzheng Wang

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Education

Santa Clara University, Leavey School of Business – Santa Clara, CA

Master of Science in Business Analytics (MSBA), Expected Graduation: 2025

Relevant Coursework: Machine Learning (Python), Data Analytics (Python), Data-Driven Marketing (R), Database Management (SQL), Prescriptive Analytics (Python), Data Visualization (Tableau)

Iowa State University – Ames, IA

Bachelor of Science in Statistics, Minor in Data Science – June 2024

Honors: Dean's List (Spring 2020)

Technical Skills

Programming: Python, R, SQL, Java

Analytics Tools: SAS, Tableau, scikit-learn, Pandas, NumPy

Data Visualization: Tableau, Matplotlib, Seaborn, ggplot2

Machine Learning & Modeling: Supervised/Unsupervised Learning (Logistic Regression, Decision Trees, Random Forest, XGBoost, LightGBM, KNN), Model Tuning (Hyperparameter Tuning, Cross-Validation), Explainable AI (SHAP)

Analytics Techniques: Clustering, Predictive Analytics, NLP (Sentiment Analysis, Topic Modeling), Data Cleaning & ETL, KPI Development

Academic Projects

Casa Grata Analytics Challenge (Energy Access)

Spring 2025 – SCU Analytics Showdown (Industry Practicum)

- Performed comprehensive SQL-based ETL on a 24-table relational dataset (~48,000 records), cleaning and joining customer, payment, and service data to enable analysis.
- Developed Guardian-level performance KPIs (e.g., payment success rate, collection volume) to evaluate and rank community agents, identifying top performers and those needing intervention.
- Analyzed payment delinquency patterns and geographic trends across rural zones to pinpoint high-risk communities and key risk factors (e.g. long payment gaps, recent inactivity), guiding targeted collection strategies.

Visualizing the Blue Economy (Barbados Project)

2024 – 2025 | Independent Data Analytics Research

- Co-authored a data-driven analysis of Barbados' Blue Economy by integrating multi-decade datasets (e.g. FAO fisheries records 1950–2022; satellite ocean metrics like sea surface temperature, productivity, Sargassum algae blooms) to examine marine-economic trends.
- Utilized Google BigQuery for data warehousing and Looker Studio dashboards to

visualize long-term changes in fisheries output, ocean climate indicators, and water clarity.

- Highlighted insights on climate impacts (rising SST, algae influx) affecting fishery yields and coastal sustainability, providing evidence-based recommendations for sustainable ocean policy planning.

Car Crash Outcome Prediction Using ML

Spring 2025 – Course: Machine Learning with Python

- Built and evaluated classification models (Logistic Regression, XGBoost, LightGBM) on 200K+ crash records to predict collision type, injury severity, and property damage extent.

- Improved model accuracy through feature engineering and hyperparameter tuning, with XGBoost achieving 54.1% accuracy in multi-class collision type prediction (best performance among models).

- Applied SHAP for explainability, revealing key risk factors (driver behavior, speeding, vehicle type) to inform targeted road safety and insurance strategies.

Hotel Revenue Growth Analysis

Fall 2024 – Course: Data-Driven Marketing (R)

- Conducted cancellation rate analysis on hotel booking data to identify trends and propose strategies for reducing lost revenue from no-shows and cancellations.

- Performed customer segmentation for loyalty program development using clustering methods, uncovering distinct guest segments and their booking behaviors.

- Applied NLP techniques (sentiment analysis and topic modeling) on guest reviews to extract actionable insights for improving hotel services and guest satisfaction.

Additional Information

Languages: English (Proficient), Chinese (Native)